VARDHAMAN COLLEGE OF ENGINEERING

(AUTONOMOUS)

Affiliated to **JNTUH**, Approved by **AICTE**, Accredited by **NAAC** with **A++** Grade, **ISO 9001:2015** Certified
Kacharam, Shamshabad, Hyderabad – 501218, Telangana, India

III B. TECH II SEMESTER

A8523–CLOUD COMPUTING AND VIRTUALIZATION LABORATORY

SYLLABUS
LIST OF EXPERIMENTS

Week-1: Create the following instances using AWS EC2

i. Linux

ii. Windows

Week-2: File transfer from Local system to AWS Linux machine and Windows machine.

Week-3: i. Launch the Windows Server from AWS and make as Web Server.

ii. Launch the Amazon Linux Server from AWS and make as Web Server.

Week-4: i. Launch the Linux machine and add 5 GB of additional volume to the Linux machine

ii. Launch the Windows Server and then add 5 GB of additional volume.

Week-5: Create and manage IAM users, Groups and Polices.

Week-6: Create and configure Amazon S3 and implement Versioning feature.

Week-7: Implement Life cycle management to the uploaded object.

Week-8: Launch the Web Server using Public subnet and Database Server using Private subnet. Connect the Database Server using Bastion Server.

Week-9: Implement and the schedule tasks with AWS Lambda.

Week-10: Connect MySQL Relational Database using Amazon RDS.

Week-11: Create tables and update in dynamo DB.

Week-12: Athena use Netflix Data Set

Week 13: Create and configure Virtual machines using Azure

Week 14: Create and manage users and groups in Azure AD, including user attributes, group memberships, and roles.

Week 15: Create, configure and manage Azure SQL Databases.

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Course Plan

Course Outcomes (COs)

After the completion of the course, the learner will be able to:

CO#	CO Statement	BL#
CO1	Create and Configure Virtual machines and Lambda Service.	BL3
CO2	Implement IAM for user management and access control	BL3
CO3	Create and manage S3 buckets with proper configurations.	BL3
CO4	Set up and Configure Virtual Private Clouds (VPCs) and Subnets.	BL3
CO5	Deploy and manage Relational Database using RDS.	BL3

Course Plan

Session #	Lecture/ Practice #	Topic	BL#	CO#
1	P1	Create the following instances using AWS EC2 i. Linux ii. Windows	BL3	CO1
2	P2	Transfer Files from Local system to AWS Linux machine and Windows machine.	BL3	CO1
3	P3	i. Launch the Windows Server from AWS and make as Web Server. ii. Launch the Amazon Linux Server from AWS and make as Web Server.	BL3	CO1
4	P4	i. Launch the Linux machine and add 5 GB of additional volume to the Linux machine ii. Launch the Windows Server and then add 5 GB of additional volume.	BL3	CO1
5	P5	Create and manage IAM users, Groups and Polices.	BL3	CO2
6	P6	Create and configure Amazon S3 and implement Versioning feature.	BL3	CO3
7	P7	Implement Life cycle management to the uploaded object.	BL3	CO3
8	P8	Launch the Web Server using Public subnet and Database Server using Private subnet. Connect the Database Server using Bastion Server.	BL3	CO4
9	P9	Implement and the schedule tasks with AWS Lambda.	BL3	CO4

Session #	Lecture/ Practice #	Topic	BL#	CO#
10	P10	Connect MySQL using RDS in AWS	BL3	CO4
11	P11	Create tables and update in dynamo DB.	BL3	CO4
12	P12	Athena use Netflix Data Set	BL3	CO5
13	P13	Create and configure Virtual machines using Azure	BL3	CO1
14	P14	Create and manage users and groups in Azure AD, including user attributes, group memberships, and roles.	BL3	CO2
15	P15	Create, configure and manage Azure SQL Databases.	BL3	CO5

Week-1	**Create the following Instances using EC2 Instance** **i. Linux Server** **ii. Windows Server**

Pre-Lab Questions	1. What is EC2? 2. What is an Instant? 3. What is the purpose of Security Group? 4. What is Key pair? **5.** List out the Operating Systems are supported by EC2.

EC2 – Introduction

EC2 stands for Amazon Elastic Compute Cloud. It is a web service that provides secure, scalable compute capacity in the cloud. It is a virtual server in Amazon's Elastic Compute Cloud (EC2) for running applications on the Amazon Web Services (AWS) infrastructure.

Amazon EC2 currently supports a variety of operating systems including: Amazon Linux, Ubuntu, Windows Server, Red Hat Enterprise Linux, SUSE Linux Enterprise Server, openSUSE Leap, Fedora, Fedora CoreOS, Debian, CentOS, Gentoo Linux, Oracle Linux, FreeBSD, and Mac OS

Security Group: A security group acts as a virtual firewall for your EC2 instances to control incoming and outgoing traffic. Inbound rules control the incoming traffic to your instance, and outbound rules control the outgoing traffic from your instance.

Key Pair: A key pair, consisting of a public key and a private key, is a set of security credentials that you use to prove your identity when connecting to an Amazon EC2 instance. Amazon EC2 stores the public key on your instance, and you store the private key.

i.Linux Server

- Select EC2 Service
- Click on Launch Instances
- Enter the Name
- Choose Amazon Machine Image (AMI) (Amazon Linux)
- Choose an Instance Type (t2.micro)
- Click on Create New Key Pair

- Enter the name
- Select Key pair type as RSA
- Select Private Key file format as .pem
- Click on create Key pair
- Enable SSH Port
- **Note: For Windows Instance RDP Port Needs to be open &**
- **for Linux Instance SSH Port**
- Select Configure storage
- Click on launch Instances
- Download Putty & Puttygen
- **Open Website**: https://www.chiark.greenend.org.uk/~sgtatham/pu_tty/latest.html
- Generate PPK File Using Puttygen
- Select the EC2 Machine & Click on connect
- Go to SSH Client
- Note down the host name
- Open putty
- Enter the Host name
- Attach PPK File with Putty
- Click on open
- Click on Accept
- Select the EC2 Machine & Click on connect
- Go to EC2 Instance Connect
- Note down the Public IPV4 Address
- Open Putty Enter IP Address

- Attach PPK File File
- Click on open
- Enter User name (ec2-user)

ii. Windows Server

- Select Region
- Select EC2 Service
- Click on Launch Instances
- Enter the Name
- Choose Amazon Machine Image (AMI) (Windows)
- Choose an Instance Type (t2.micro)
- Click on Create new key pair
- Enter the key pair name (**Special Character doesn't work**)
- select Private Key file format as .pem
- Click on create key pair
- Check Allow RDP Traffic from
- Select source as Anywhere
- Select Configure storage
- Click on Launch Instances
- Select the Instance

- Click on Stop/ Terminate

Week-2	Transfer Files from Local System to Launched Linux Server and Windows Server

Pre-Lab Questions	1. Define Server? 2. What is an Instant? 3. What is the purpose of Security Group? 4. What is Key pair? 5. List out the Operating Systems are supported by EC2.

Transfer Local Files to Windows Server

- Create Windows EC2 Machine
- Generate the password
- Create File in Local Machine
- Open Remote Desktop Connection
- Click on Show Options
- Go to Local Resources
- Click on More
- Check the Drivers
- Click on ok
- Click on Connect
- Now Copy the File & Paste the File in the EC2 Machine

Transfer Local Files to Linux Server

- Create Linux EC2 Machine
- Generate the PPK File
- Download & Install WINSCP:

  https://winscp.net/eng/download.php
- Enter the Host Name & User Name

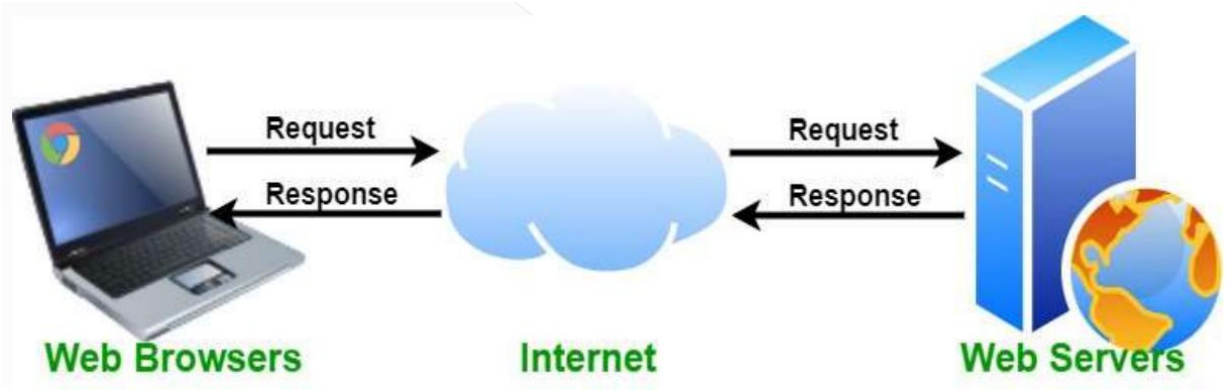
- Attach PPK File
- Click on ok
- Click on Login
- Click on Yes
- Drag and Drop the file in Linux server
- Now we can connect to Putty & Check using Linux Commands.

Week-3	**Launch the Windows Server from AWS and make as Web Server.** **Launch the Amazon Linux Server from AWS and make as Web Server.**

Pre-Lab Questions	1. What is Web Server? 2. What is an Instant? 3. What is Repository? 4. What is YUM Repository? **5.** How to Create empty files in Linux?

Launch the Amazon Linux Server from AWS and make as Web Server.

A Web server is a program that uses HTTP (Hypertext Transfer Protocol) to serve the files that from Web pages to users, in response to their request.



Example of Webservers

- Apache HTTP Server
- Internet Information Services (IIS)
- NGINX
- HTTPD by Apache

HTTPD: HTTP Daemon is a software program that runs in the background as a web server. It waits for the incoming server requests. The daemon answers the request and serves the hypertext and multimedia documents over the Internet using HTTP.

Linux Web Server:

We will install HTTPD by Apache & it will be the Two Step Process

Step 1: Update the YUM Repository

Step 2: Install Web Package

- Create Linux EC2 Machine
- Connect the Linux EC2 Machine
- Switch User to root

  sudo su
- Update the YUM Repository

  yum update –y
- Install Web Package

  yum install httpd –y
- Move to the html Directory

  cd /var/www/html
- Create web Page

  echo “Amazon Web Services” > index.html
- Start web package

  service httpd

  start chkconfig httpd on
- Now, Our Linux machine is webserver.
- Copy the Public IP & paste in Web Browser
- Select our Security group
- Edit Inbound Rules
- Click on Add Rule.
- Change type to HTTP & Source as Anywhere IPV4
- Click on Save rules
- Now Refresh the Web browser

Repository: A Linux repository is a storage location from which your system retrieves and installs OS updates and applications.

We need to install web package, it is called web server. Package is nothing but software.

YUM Repository: YUM Repositories are warehouses of Linux software. Sometimes the software we want to install is not available in Linux OS default. In such situations, we can use YUM Repositories. We can install new software on Linux with “yum install packagename" command from console.

Launch the Windows Server from AWS and make as Web Server.

- Create One Windows EC2 Machine
- Generate the password
- Connect the EC2 Machine using the RDP
- Click on Start Menu
- Open Server Manager
- Install Internet Information Services (IIS)
- Click on Add roles and features
- Click on ok
- Click on Next
- Click on Next
- Click on Next
- Check Web Server (IIS)
- Click on Add Features
- Click on Next
- Click on Next
- Click on Next
- Click on Next
- Click on Install
- Check localhost
- Create own page
- Go to the path C:\inetpub\wwwroot
- Create Index.html file in the same path
- HTML Code:

  </h1>Amazon Web Services</h1>

- Enable HTTP Port

- Click on Security Group ID
- Click on Edit inbound rules
- Click on Add rule
- Select type as HTTP & Source Anywhere IPV4
- Click on save rules
- Copy the Public IP and Paste in browser

Week-4	**Launch the Linux Server and 5GB of additional Volume to the Linux Server** **Launch the Windows Server and then add 5GB of additional Volume**

Pre-Lab Questions	1. What is Elastic Block Store? 2. How to attach and detach the additional Volume Linux Server? 3. What is Repository? 4. What is YUM Repository? **5.** How to Create empty files in Linux?

Launch the Linux Server and then add 5GB of additional Volume

- One EC2 Machine can have multiple Volumes also & Additional volume is known as Elastic Block Store (EBS)
- Creating Linux EC2 Machine using the multiple Volumes
- Go to Add Storage
- Click on Add New Volume
- Check Delete on Termination
- Launch the Instance
- Check Volumes

- Mount the volume
- Connect Linux EC2 machine using putty
- Check disk space usage

  df -hT
- Check available disk device

  lsblk

Note down the name of the added ebs volume (xvdb)

- Information about device

sudo file -s /dev/xvdb

- Create file system on the volume

  sudo mkfs -t ext4 /dev/xvdb

- Create Directory

  mkdir test

- Mount the Volume

  sudo mount /dev/xvdb test

- Change directory

  cd test

- Create files

  sudo touch file1 file2 file3

- Check the files

  ls

- Check disk space usage

  df –hT

- Change directory

  cd

- Unmount the EBS volume

  sudo umount test

- Detach Volume

- Click on Detach
- Create another Linux ec2 machine with root volume in same availability zone in which we have our EBS volume
- Attach Volume
- Select the Volume which we want to attach with EC2 Machine

- Select the instance Click on attach volume
- Now connect the second ec2 machine using putty
- Check disk space usage

  df –hT
- Check available disk device

  lsblk

Note down the name of the added ebs volume (xvdf)

- Information about device

  sudo file -s /dev/xvdf
- Create directory

  mkdir training
- Mount the Volume

  sudo mount /dev/xvdf training
- Change directory

  cd training
- Check the files

  ls

Launch the Windows Server and then add 5GB of additional Volume

- Create Windows EC2 Machine with EBS Volume
- Check the volumes

- Connect the instance using remote desktop connection Click on start button
- Open server manager
- Click on files and storage services
- Click on disks
- Select the disk & right click
- Click on bring online
- Click on yes
- Right click on the disk

- Click on initialize
- Click on yes
- Right click on the disk
- Click on new volume
- Click on next (Multiple Times)
- Click on create
- Now check the windows system
- Open D Drive & create one file
- Unmount the volume from ec2 machine
- Go to Server manager
- Click on files and storage services
- Click on disks
- Right click on EBS disk
- Click on Take Offline
- Click on Yes
- Detach the volume
- Select the volume

- Click on Detach
- Create windows ec2 machine in the same availability zone in which we have our EBS volume
- Attach the EBS volume with instance

- Select the instance Click on attach volume
- Connect the new windows ec2 machine using Remote desktop connection
- Open server manager
- Click on file and storage services
- Click on disks
- Our ebs disk is showing as offline.
- Make that disk online

- Select the ebs disk
- Right click on ebs disk & click on being online
- Click on Yes
- Now check this PC
- Open D Drive

Week-5	Create and manage IAM Users, Groups and Policies.

Pre-Lab Questions	1. What is Identity and Access Management (IAM)? 2. What are the key features of AWS IAM? 3. Define AWS users, groups, and roles? 4. What is MFA in AWS IAM? 5. What are IAM users' access and secret keys?

AWS Identity and Access Management (IAM) enables you to manage access to AWS services and resources securely. Using IAM, you can create and manage AWS users and groups, and use permissions to allow and deny their access to AWS resources. IAM can also keep our account credentials private.

When we first create an AWS account, it has complete access to all AWS services. This identity is called the AWS account root user

IAM Features:

Shared access to the AWS account: The main feature of IAM is that it allows you to create separate usernames and passwords for individual users or resources and delegate access.

Multifactor authentication (MFA): IAM supports MFA, in which users provide their username and password plus a one-time password from their phone a randomly generated number used as an additional authentication factor.

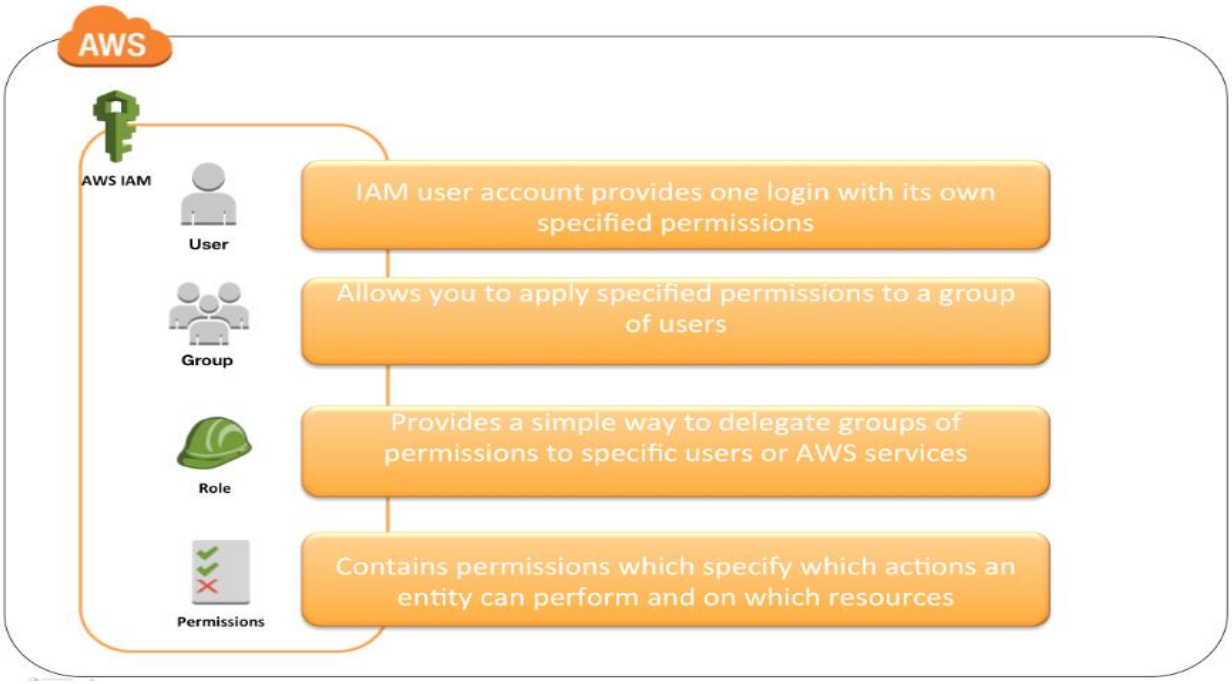
Identity Federation: If the user is already authenticated, such as through a Facebook or Google account, IAM can be made to trust that authentication method and then allow access based on it.

Free to use: There is no additional charge for IAM security. There is no additional charge for creating additional users, groups or policies.

Password policy: The IAM password policy allows you to reset a password or rotate passwords remotely.

Granular permissions: Each user can be granted with different set granular permissions as required to perform their job

AWS IAM Identities:



AWS is supporting two types of accounts

1. **Root User**
2. **IAM User**

Root User:

- Root Account Credentials are the email address and password with which we sign in into the AWS account
- Root Credentials has full unrestricted access to AWS account including the account security credentials which include sensitive information
- An Administrator account can be created for all the activities which too has full access to the AWS account except the accounts security credentials, billing information and ability to change password

IAM User:

- IAM user represents the person or service who uses the access to interact with AWS.
- IAM user starts with no permissions and is not authorized to perform any AWS actions on any AWS resources and should be granted permissions as per the job function requirement.

- Each IAM user is associated with one and only one AWS account.
- IAM User cannot be renamed from AWS management console and has to be done from CLI or SDK tools.

Steps to Create IAM User:

- Create Two IAM user.
- One user will access only EC2 Machines & Second user will access only S3 Buckets.
- Go to IAM
- Click on Users
- Click on Add Users
- Enter the user name
- Select AWS Access Type.
- We can connect our AWS account with 2 ways.
- Console Access (Graphical Access)
- Command Line Interface (CLI)
- Select custom password & enter the password
- Uncheck require password reset
- Click on Next: Permissions
- Click on Attach existing policies directly
- Search the EC2 full Access policy
- Select the policy
- Click on Next: Tags
- Click on Next: Review
- Click on Create user
- Create Another User Given S3 Full Access
- Note down the console ID of your root user
- Go to the AWS URL
- Select IAM User
- Enter the Account ID (Console ID)
- Click on Next

- Enter IAM user name & password
- Click on Sign In
- Change the Password

IAM Groups and Attach Policies:

- An IAM Group is a collection of users. Group specifies the permission for a collection of users, and it also makes it possible to manage the permissions easily for those users.
- Following are some important characteristics of user groups:
- A user group can contain many users, and a user can belong to multiple user groups.
- User groups can't be nested; they can contain only users, not other user groups.
- There is no default user group that automatically includes all users in the AWS account. If you want to have a user group like that, you must create it and assign each new user to it.
- Default Quota is 300 Groups For an AWS Account & Max quota is 500 Groups

Steps to Create Groups and attach Policies:

- Go to IAM
- Click on User groups
- Click on Create Group
- Enter the name of the group
- Attach Permission policies
- Search S3Full Access
- Click on Create Group
- Create new user with Group
- Select AWS Access Type
- Enter password
- Click on Next: Permissions
- Select Add user to group
- Click on Add: Tags
- Click on Create User
- Login with User

- Check the S3 Bucket
- Remove User & Add User in Group
- Create One more Group
- Attach Permissions Policies
- Search EC2 Read only
- Create Group
- Open First Group
- Select the user
- Click on Remove users
- Open Second Group
- Click on Add Users
- Select the user & Click on Add users
- Now check group

Week-6	Create and Configure Amazon S3 and Implement Versioning Feature.

Pre-Lab Questions	6. What is the full form of S3? 7. List out the features of S3? 8. Naming conventions of creating the bucket? 9. What is versioning? 10. What are the advantages of Versioning?

Simple Storage Service (S3)

S3 is a secure, durable and highly scalable object storage. S3 is easy to use, with a simple web service interface to store and retrieve any amount of data from anywhere on the web. This means customers of all sizes and industries can use it to store and protect any amount of data for a range of use cases, such as data lakes, websites, mobile applications, backup and restore, archive, enterprise applications, IoT devices, and big data analytics.

S3 Features:

- It is a global service
- Low latency and high throughput performance
- It is a Safe place to store your files
- It is object based storage
- Durability: AWS claims Amazon S3 to have a 99.999999999% of durability. This means the possibility of losing your data stored on S3 is one in a billion.

- A single Amazon S3 object can store maximum of 5 terabytes. The total volume that an s3 bucket can store is unlimited. If an object size is greater than 5 gigabytes, you should consider multipart upload.
- S3 Bucket Names must be unique globally.
- Lifecycle Management
- Versioning
- Encryption
- Secure Your data using Access Control lists & Bucket Policy

S3 Restrictions and Limitations:

- Bucket ownership is not transferable to another account.
- When we create a bucket, we choose its name and the AWS Region to create it in. After we create a bucket, we can't change its name or Region.
- By default, we can create up to 100 buckets in each of our AWS accounts. If we need additional buckets, you can increase our account bucket limit to a maximum of 1,000 buckets by submitting a service limit increase. There is no difference in performance whether you use many buckets or just a few. (Note: Search Service Quotas in AWS)

S3 Bucket Overview:

- Amazon S3 Allows people to store objects (Files) in the “buckets” (Directories)
- Buckets must have a globally unique names
- Buckets are defined at the region level
- Naming Convention

  No Uppercase

  No Underscore

  3 – 63 Characters long

  Not an IP Address (224.245.278.135)

  Must start with lowercase letter or number

S3 Create Bucket:

- Click on Create Bucket
- Enter the bucket name

- Select the region
- Select ACLs Enabled
- Uncheck the block all public access
- Check the acknowledge
- Click on Create bucket

S3 Upload Object:

- Click on the name of the bucket
- Click on Upload
- Click on Add files
- Select the file which we want to upload in our S3 bucket
- Click on permissions
- Select grant public read access
- Click on Upload

S3 View Object:

- Click on the name of the object
- Go to the properties tab
- Copy the object URL
- Paste in the browser
- Change permission of the object
- Click on the name of the object
- Go to Permission tab
- Click on Edit
- Uncheck read access from public access
- Click on save changes
- Paste the object URL in browser

Versioning:

Versioning in Amazon S3 is a means of keeping multiple variants of an object in the same bucket. We can use the S3 Versioning feature to preserve, retrieve, and restore every version of

every object stored in our buckets. It is enabled at the bucket level. Versioning- enabled buckets can help you recover objects from accidental deletion or overwrite.

Advantages of Versioning:

- We can recover deleted object.
- We can maintain different versions of the file

Steps for Versioning:

- Create bucket
- Select ACLs Enabled
- Don’t give access to public
- Click on Create bucket
- Upload the Object
- Don’t give any permission
- Paste the Object URL in Browser
- Give Public Access to Bucket
- Open our bucket
- Go to Permissions tab
- Click on Edit for block public access
- Uncheck the Block all public Access
- Click on save changes
- Enter confirm & Click on Confirm
- Now open File
- Go to Permission tab.
- Click on edit
- Check the read check box for public
- Check the authentication
- Click on save changes
- Check the object URL

Versioning Recover Deleted Object:

- Open our Bucket

- Go to properties tab
- Click on Edit for bucket Versioning
- Select enable
- Click on save changes
- Open our bucket
- Click on show versions
- Delete the file
- Select the file & Click on Delete
- Enter delete
- Click on delete objects
- Click on show versions
- Select delete marker file & Click on delete
- Enter permanently delete
- Click on Delete objects

Versioning Maintain Different Versions:

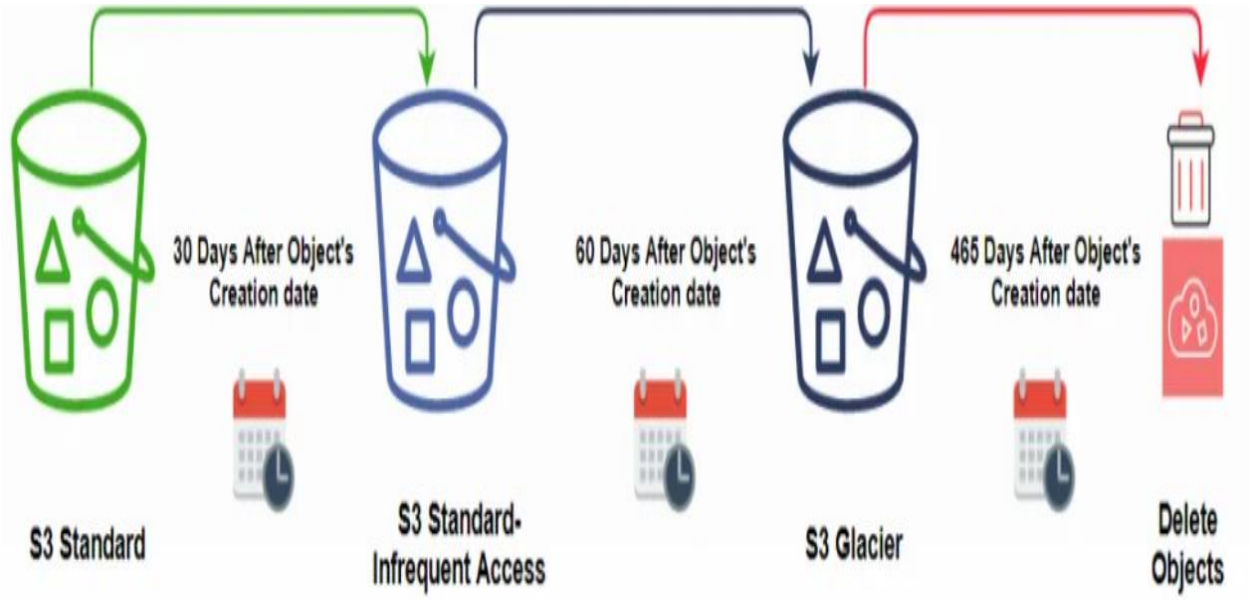
- Add more lines in our existing file
- Open bucket & Upload the new file
- Go to Permissions
- Give Public Access
- Click on upload
- Click on show versions
- Recover Old Version File
- Select the old version file
- Click on download
- System will download the file & if we want then we can upload the file again.

Week-7	Implement Life Cycle management to upload the object.

Pre-Lab Questions	1. What is the Maximum size of the object can stored in bucket? 2. What are different Storage Classes in Amazon S3? 3. How can you secure data in S3? 4. What tools does AWS offer for monitoring S3? 5. What are ACLs in AWS S3?

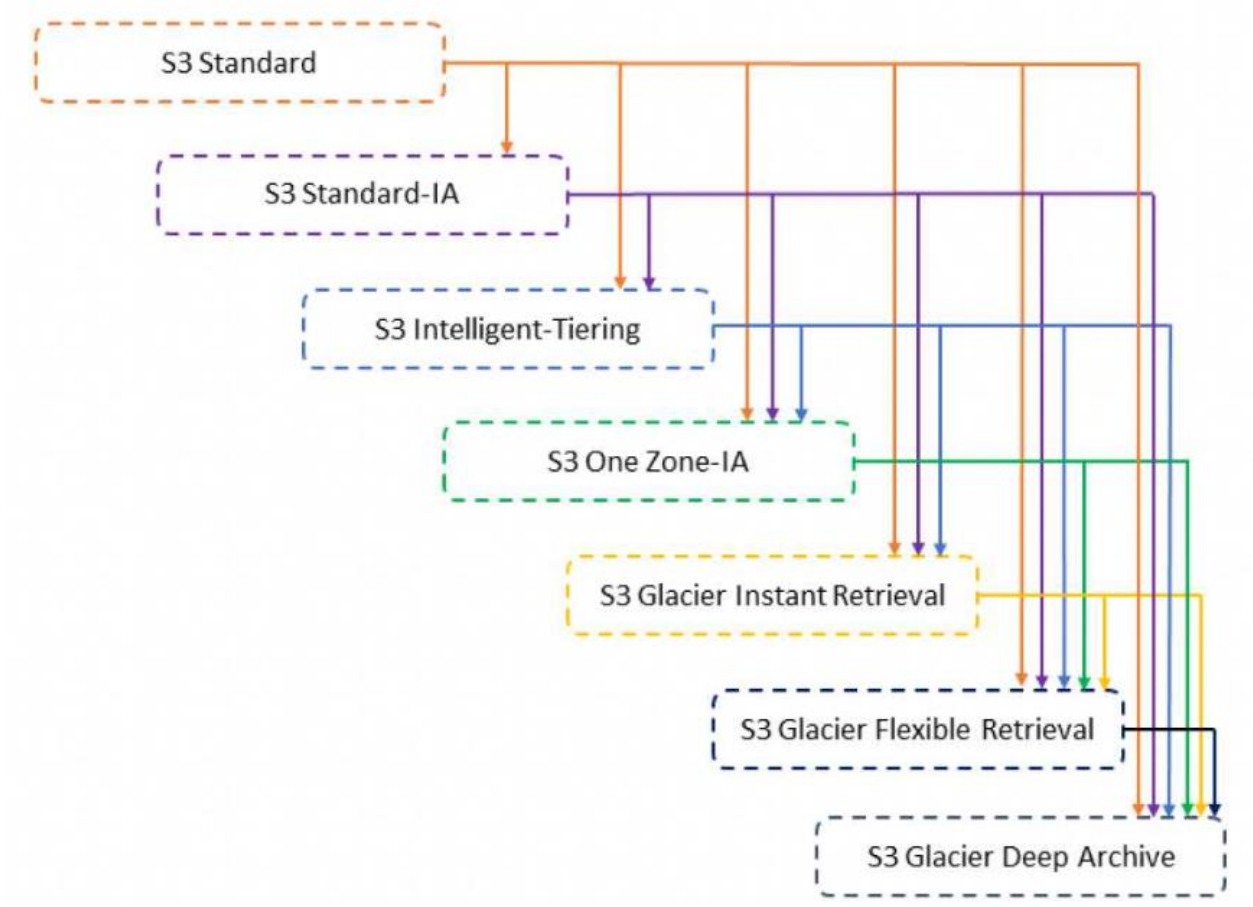
Life Cycle Management:

- It is a way of efficiently managing the objects so that the cost is minimized throughout the lifecycle of the lifecycle of the object inside the instance. 1000 lifecycle rules can be configured per bucket .
- There are two types of actions:
- Transition actions: in which the storage class for the objects changes
- Expiration actions: where the objects expire and are permanently deleted



S3 Lifecycle Configuration Example

S3 Object Lifecycle Management Rules



Steps to apply to Life Cycle Management:

- Create Bucket
- Select ACLs Enabled
- Give Public Access
- Upload the object
- Give the public Access
- Open bucket
- Go to Management tab
- Click on Create lifecycle rule
- Enter the Name
- Select this rule applies to all objects in the bucket
- Acknowledge the rule

- Select the rules as per our client requirements
- Select storage classes
- Select the class & enter the days
- Check timeline summary
- Click on create rule

Week-8	Launch the Web Server using Public Subnet and Database Server using Private Subnet. Connect the Database Server using Bastion Server.

Pre-Lab Questions	1. What is VPC? 2. What are the Components of AWS VPC? 3. What is Internet Gateway> 4. What is NAT Gateway? 5. What is Subnet in VPC? 6. Difference between Private, Public and Elastic IP Addresses? 7. What is CIDR full form?

Virtual Private Cloud:

A virtual private cloud (VPC) is a virtual network dedicated to your AWS account. It is logically isolated from other virtual networks in the AWS Cloud. You can launch your AWS resources, such as Amazon EC2 instances, into your VPC. There are no additional charges for creating and using an Amazon Virtual Private Cloud (VPC) itself.

Steps for Creating VPC:

- Search VPC
- Click on Create VPC
- Enter the Name of the VPC
- Enter IPV4 CIDR
- Select tenancy as default.
- Click on Create VPC

Subnet:

A subnet is a segment of an Amazon VPC's IP address range where you can launch Amazon EC2 instances, Amazon Relational Database Service (Amazon RDS) databases, and other AWS resources. We can add one or more subnets in each Availability Zone.

Subnet Types:

Private Subnet: Instances in the private subnet are back-end servers that don't need to accept incoming traffic from the internet and therefore do not have public IP addresses. By default every subnet is the private subnet. For Example Database server

Public Subnet: A public subnet is a subnet that's associated with a route table that has a route to an internet gateway. For Example Webserver

Steps for Creating Subnet:

- Click on subnets
- We are able to see all the default subnets
- Click on create subnet
- Select the VPC in which we want to create the subnet
- Enter subnet name, Select availability zone & Enter IPV4 CIDR Block.
- Click on Create subnet
- Create One More Subnet

MAKE SUBNET AS PUBLIC:

- Select Subnet
- Actions → Edit subnet
- Check the Enable auto-assign public IPV4 address
- Click on save.

Internet Gateway:

Component that allows communication between our VPC and the internet

Steps for Creating Internet Gateway:

- Click on internet Gateway
- Click on Create internet gateway
- Enter the name of the Internet gateway
- Click on Create Internet gateway
- Attach internet gateway to our VPC
- Select our internet gateway
- Actions → Attach VPC
- Select our VPC
- Click on Attach internet gateway
- We need to attach our internet gateway to our Public Subnet. But we can’t attach internet gateway directly to the public subnet.

Route Table:

Route table defines rules, termed as routes, which determine where network traffic from the subnet would be routed.

Steps for Creating Route Table:

- Select our route table
- Actions → Edit Subnet Association
- Select the public subnet
- Click on save Associations
- Attach Route table with Internet gateway
- While attaching the route table with internet gateway we need to provide two elements
- Destination
- Target
- Select the route table
- Actions → Edit Routes
- Click on Add route
- Select target as our Internet gateway
- Enter Destination (0.0.0.0/0)
- Click on save changes

EC2 – Public Subnet:

- Create Linux EC2 Machine in Public Subnet
- Change network, Subnet & Auto Assign Public IP
- Bootstrap Script Code:

```
#!/bin/bash
sudo su
yum update -y
yum install httpd -y
cd /var/www/html
echo "AmazonWebservices" > index.html
service httpd start
chkconfig httpd on
```

- Create Security Group
- Enable the SSH Port to anywhere as source

- Enable the HTTP Port to anywhere as source
- Launch the Instance & Create a new Key Pair

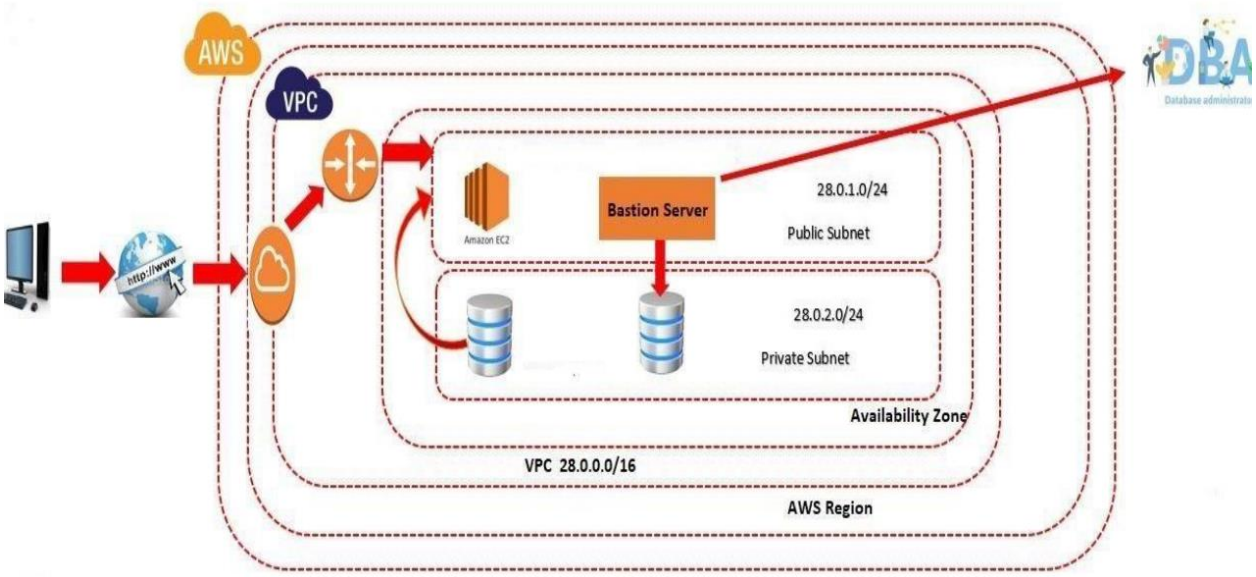
EC2 – Private Subnet

- Create One More Linux EC2 Machine in Private Subnet
- Select Our VPC & Private Subnet
- Create Security Group
- Select Type as MYSQL/Aurora & Source as Custom
- Enter Port Number of Public Subnet
- Launch the Instance
- Create a new Key Pair

Bastion Server:

A bastion host is a server whose purpose is to provide access to a private network from an external network, such as the Internet.

Note: Bastion server is nothing but normal EC2 Machine.



- Create One More Linux EC2 Machine in Public Subnet
- Create Security Group
- Enable SSH Port Number
- Select Source as MY IP
- Launch the Instance

- Create a new keypair
- Note down the Private IP Address of Bastion Server
- Select our Database server & go to Security tab
- Click on our security group
- Actions →Edit Subnet Rules
- Click on Add Rule
- Select type as SSH & Source as Custom
- Enter the bastion server private IP
- Connect to database server using the bastion server
- Connect Bastion Server
- Switch user

  sudo su
- Update repository

  yum update -y
- Copy SSH Command of database server & paste in putty
- Enter yes
- Transfer Database server pem file to Bastion server using WINSCP
- Copy SSH Command of database server & paste in putty
- chmod 600 .pem • Switch User sudo su
- Command to install MYSQL database

  yum install mysql -y
- We cannot install, as we are not having internet connection to private subnet.
- To get internet connection, we create NAT server. (Network Address Translation)

Week-9	Implement and Schedule tasks with AWS LAMBDA

Pre-Lab Questions	1. What is LAMBDA Service? 2. What are the features in AWS lambda that automate the deployment? 3. What are the various ways to access EC2? 4. Tell us about the frameworks which are available for serverless? 5. What are the advantages of using the Serverless approach? 6. What makes Lambda a time-saving approach?

LAMBDA:

AWS Lambda is a service which computes the code without any server. It is said to be server less compute. The code is executed based on the response of events in AWS services.

To get working with AWS Lambda, we just have to push the code in AWS Lambda service. All other tasks and resources such as infrastructure, operating system, maintenance of server, code monitoring, logs and security is taken care by AWS.

Lambda Function support Code:

- Node.js
- Python
- Ruby
- Java
- Go
- C#
- Powershell

Advantages of Lambda:

- Ease of Working with Code
- Logs Provision
- Billing based on Usage
- Multi Language Support

LAMBDA - INTEGRATION WITH OTHER AWS SERVICES:

- AWS API Gateway
- Amazon S3

- AWS Cloud Formation
- AWS Kinesis
- AWS CloudWatch Events
- AWS DynamoDB
- AWS SNS etc

Steps for Creating Lambda:

1. **Create IAM Role**

- Create IAM Role
- Go to IAM Service
- Click on Roles
- Click on Create Role
- Select AWS Services
- Select Lambda & Click on Next: Permissions
- Search AmazonDynamoDBFullAcces
- Enter the Role Name
- Click on Create Role

2. **Create S3 Bucket:**

   Unblock all public access and acknowledge the settings

3. **Create DynamoDB Table:**

   - Enter the table Name (newtable)
   - Enter partition key (unique)
   - Click on Create table

4. **Create Lambda Function**

   - Go to Lambda Service
   - Click on Create Function
   - Enter the Function Name
   - Select Runtime (Python 3.6)
   - Click on Change default execution role
   - Select the use an existing role
   - Select the role we have created

- Click on Create function
- Lambda Python Code:

```
import boto3
from uuid import uuid4
def lambda_handler(event, context):
   s3 = boto3.client("s3")
    dynamodb = boto3.resource('dynamodb')
    for record in event['Records']:
        bucket_name = record['s3']['bucket']['name']
        object_key = record['s3']['object']['key']
        size = record['s3']['object'].get('size', -1)
      event_name = record ['eventName']
        event_time = record['eventTime']
        dynamoTable = dynamodb.Table('newtable')
        dynamoTable.put_item(
            Item={'unique': str(uuid4()), 'Bucket':
bucket_name, 'Object': object_key,'Size': size, 'Event':
event_name, 'EventTime': event_time})
```

- Click on Deploy
- Click on Add Trigger
- Select trigger as S3
- Select the bucket
- Select event type as per requirement (All Object Create Events)
- Check the acknowledge
- Click on Add
- Upload the Object in S3 Bucket & Give Public Access to Object
- Now check the DynamoDB table
- Click on Run

Week-10	Connect MySQL Using Amazon RDS

Pre-Lab Questions	1. What is RDS? 2. What is the difference between Amazon RDS and Amazon EC2? 3. What is an RDS Interface? 4. What is the impact of creating an RDS Snapshot on performance? 5. What are the advantages of AWS RDS? 6. What are the differences between RDS, DynamoDB and Redshift?

Amazon RDS supports an array of database engines to store and organize data. It also helps with relational database management tasks, such as data migration, backup, recovery and patching.

Advantage of RDS:

- Automatic Backups
- Multi Availability Zone
- Read Replica

RDS – BACKUP:

Backup refers to the copying of physical or databases to a secondary location for preservation in case of equipment failure.

Types of Backups

1. Automatic Backup
2. Manual Backup

Steps to follow to Connect MySQL using RDS:

- Click on RDS
- Click on create database
- Select standard create as database creation method
- Select MySQL (It is open source database & it is given as free trail)
- Select free tier
- Enter the database name
- Enter Master password & confirm password
- Database instance class

- Go to Storage
- Go to Availability & Durability
- Go to Connectivity
- Select default VPC & Select default subnet group
- Select Yes for Public Access
- Select Create new for Security group & enter the name for security group
- Select availability Zone
- Click on Additional Configuration
- Automatic Backup
- Click on Create database
- Download MYSQL Workbench:

  MySQL :: Download MySQL Workbench
- Click on Database identifier
- Note down the endpoint & port number
- Open MySQL Workbench
- Click on Database
- Click on Connect to database
- Enter hostname
- Enter username (admin)
- Click on store in vault
- Enter password
- Click on ok
- SQL Query:

```
-- Create a database
-- DROP DATABASE IF EXISTS quickstartdb;
CREATE DATABASE quickstartdb;
USE quickstartdb;

-- Create a table and insert rows
DROP TABLE IF EXISTS inventory;
CREATE TABLE inventory (id serial PRIMARY KEY, name VARCHAR(50), quantity
INTEGER);
```

```
INSERT INTO inventory (name, quantity) VALUES ('banana', 150);
INSERT INTO inventory (name, quantity) VALUES ('orange', 154);
INSERT INTO inventory (name, quantity) VALUES ('apple', 100);

-- Read
SELECT * FROM inventory;

-- Update
UPDATE inventory SET quantity = 200 WHERE id = 1;
SELECT * FROM inventory;

-- Delete
DELETE FROM inventory WHERE id = 2;
SELECT * FROM inventory;
```

RDS – DELETE THE DATABASE

- Select the Instance
- Uncheck create final snapshot
- Acknowledge the deletion
- Enter delete me
- Click on Delete

Week-11	Create DynamoDB Table and Update it

Pre-Lab Questions	1. What is DynamoDB? 2. What is the purpose of DynamoDB? 3. What type of data can be stored in DynamoDB? 4. What is the partition key in DynamoDB? 5. What is the sort key in DynamoDB? 6. What are the features of DynamoDB?

Amazon DynamoDB is a fast, fully-managed NoSQL database service that makes it simple and costeffective to store and retrieve any amount of data, and serve any level of request traffic.

Features of DynamoDB:

DynamoDB has Two Constraints

1. Primary Key (Partition Key)
2. Composite Primary Key (Sort Key)

Steps to Create DynamoDB Table:

- Go to DynamoDB
- Click on Create table
- Enter the table Name
- Enter Partition Key
- Enter Sort Key
- Select the default settings
- Click on Create table
- Click on Name
- Click on Explore table items
- Click on Create item
- Enter the value
- Click on Add New Attributes for more columns
- Select type as string
- Enter the attribute name & value
- Click on Create Item

- We can create more items
- To delete dynamoDB Table, Select table and then Click on delete.

Week-12	Athena Using Netflix Data Set

Pre-Lab Questions	1. What is DynamoDB? 2. What is the purpose of DynamoDB? 3. What type of data can be stored in DynamoDB? 4. What is the partition key in DynamoDB? 5. What is the sort key in DynamoDB? 6. What are the features of DynamoDB?

Introduction to AWS Athena *Amazon Athena* is an interactive query service that makes it easy to analyze data directly in Amazon Simple Storage Service (Amazon S3) using standard SQL.

Features:

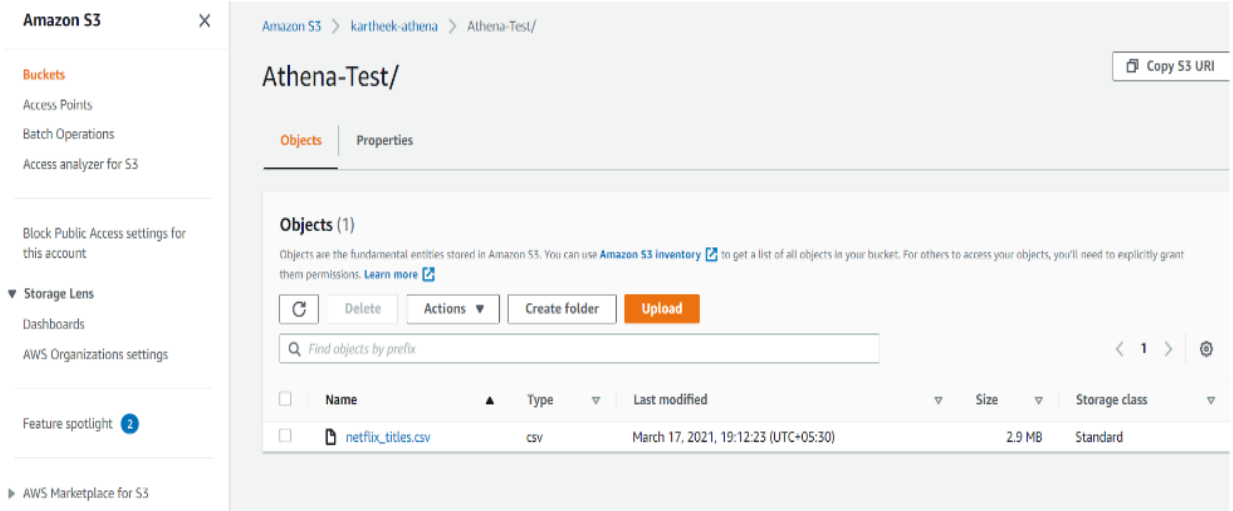
- Athena is serverless. So there is no infrastructure to set up or manage, and you pay only for the queries you run.
- It scales automatically and also executes queries in parallel.
- Results are fast even with large data sets and complex queries.
- Athena helps you to analyze unstructured, semi-structured, and structured data stored in amazon s3. eg: Apache Web Logs, CSV, TSV, Text file with custom delimiters, JSON, etc.
- We can access Athena using AWS management console, JDBC or ODBC connection, Athena API, Athena CLI.
- It is Cost-effective – We pay only for S3 which is quite cheap, and externally we pay per query. Per query cost – 5 dollars per Terabyte scan for S3.

Presto (SQL on anything): Athena uses Presto as a managed service. Presto is an in-memory distributed SQL engine, which came out of Facebook. It reads data from anywhere and actually processes data from where it lives; hence it can be connected to a variety of connectors including HDFS, S3, MongoDB, MySQL, Postgres, Redshift, SQL Server. It has the power to handle hundreds of concurrent queries on a single cluster. That means we don't have to maintain complex clusters.

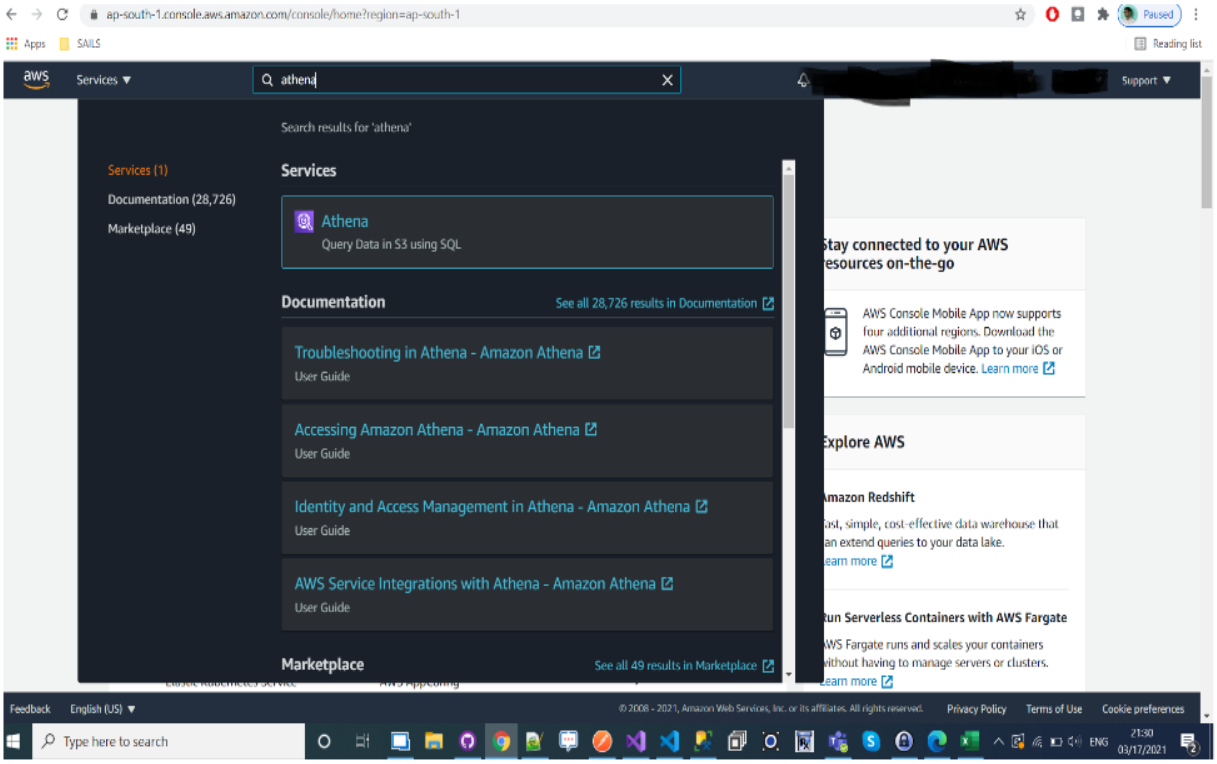
So it supports commands like creating a table, nested queries, multiple joins. We can also partition the data based on any column, not just date and time, but can also make a combination of several columns. It uses Hive QL for DDL, and Presto while querying the data.\

Querying data from S3-Bucket using Athena:

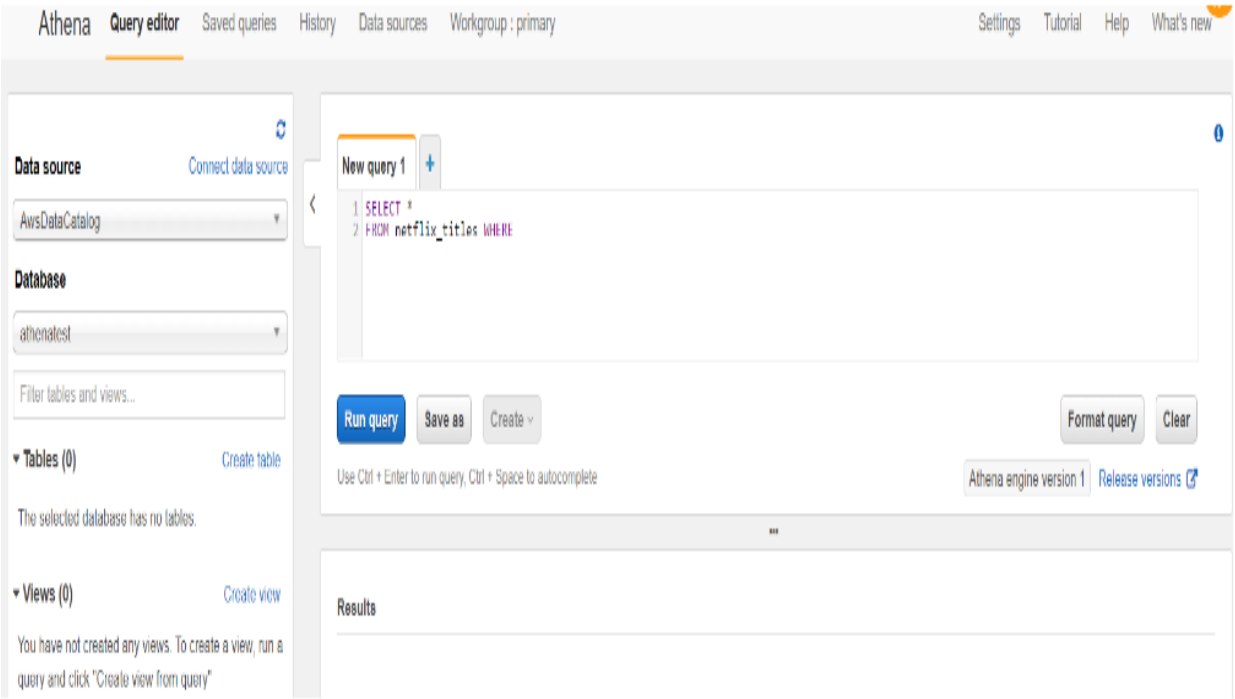
We took sample data set from **https://www.kaggle.com/shivamb/netflix-shows** which is of CSV format and uploaded it to s3://kartheek-athena/ which is test bucket.



On global search type Athena and select Athena-Query data in S3 using SQL.



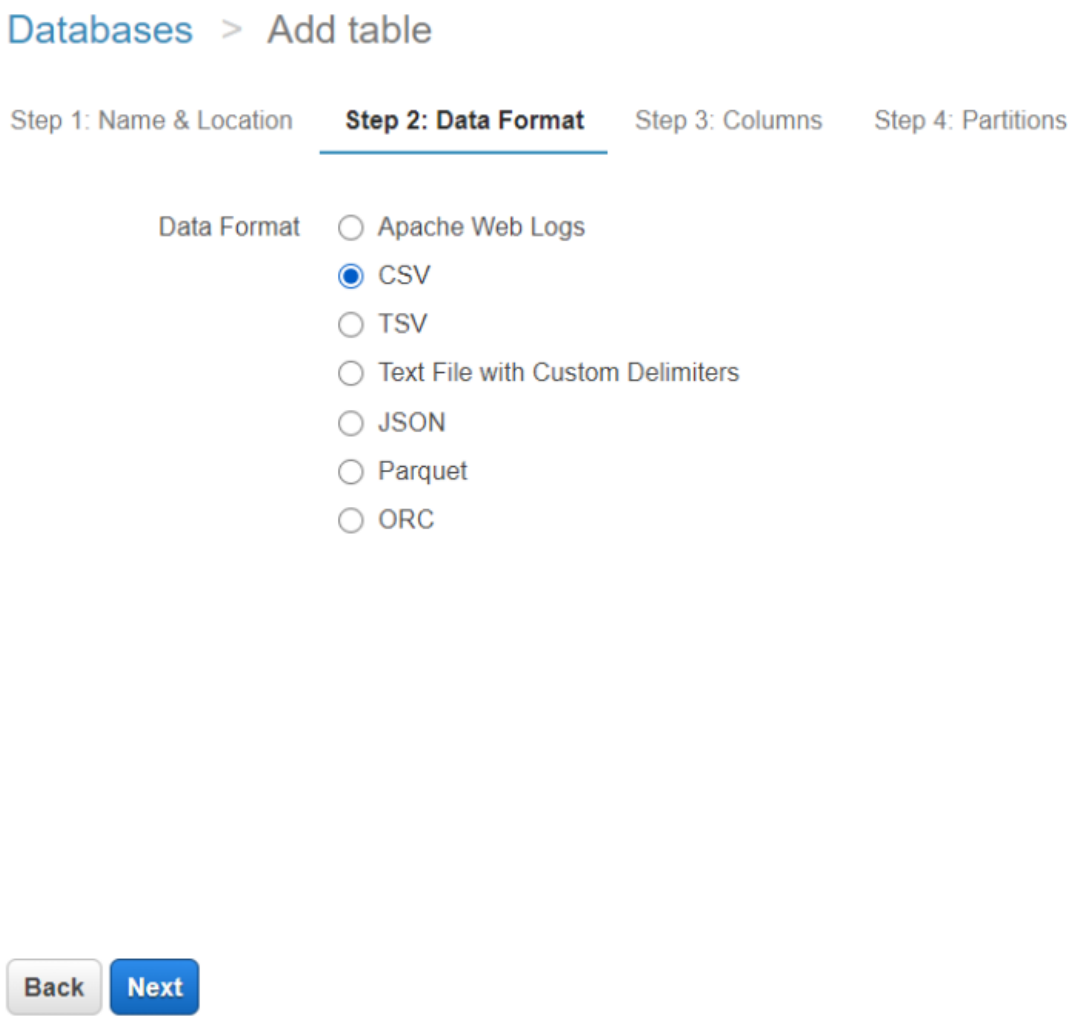
- Now you will see the following screen, we need to click on **Connect Data Source**.



- Here we need to choose where our data is located, I will select Query data in Amazon S3 and will click on the Next button.
- Now we need to specify connection details i.e Athena will connect to your data stored in Amazon S3 and use AWS Glue Data Catalog to store metadata, such as table and column names.
- This can be done in two ways i.e setup a crawler in AWS Glue to retrieve schema information

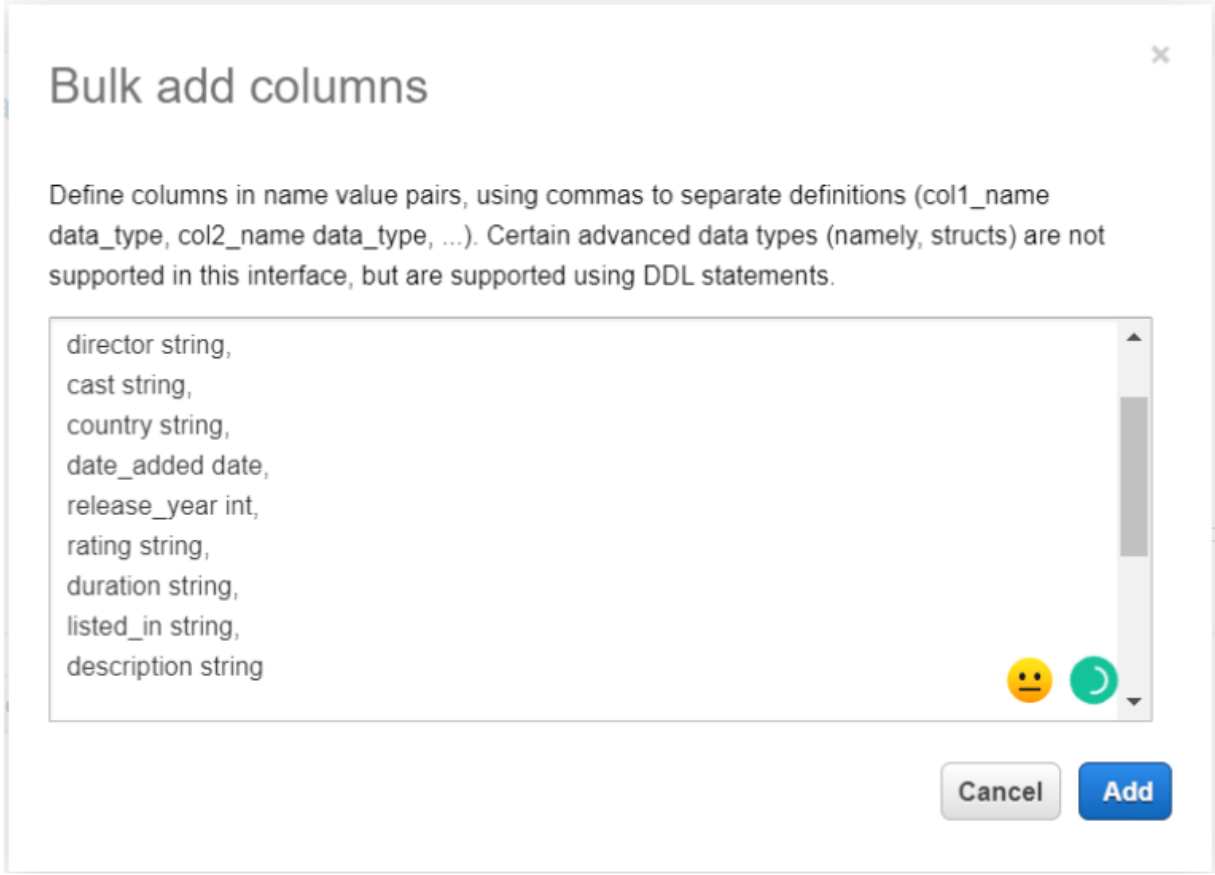
  automatically and another way is to add a table and enter schema informationmanually.
- This can be done in two ways i.e setup a crawler in AWS Glue to retrieve schema information automatically and another way is to add a table and enter schema information manually.

- Select the second option and click on the continue to add table button.
- Here we need to specify the database name, table name, and location of the input data set and click on the next button.
- Now we need to select data format i.e CSV and click on the next button.

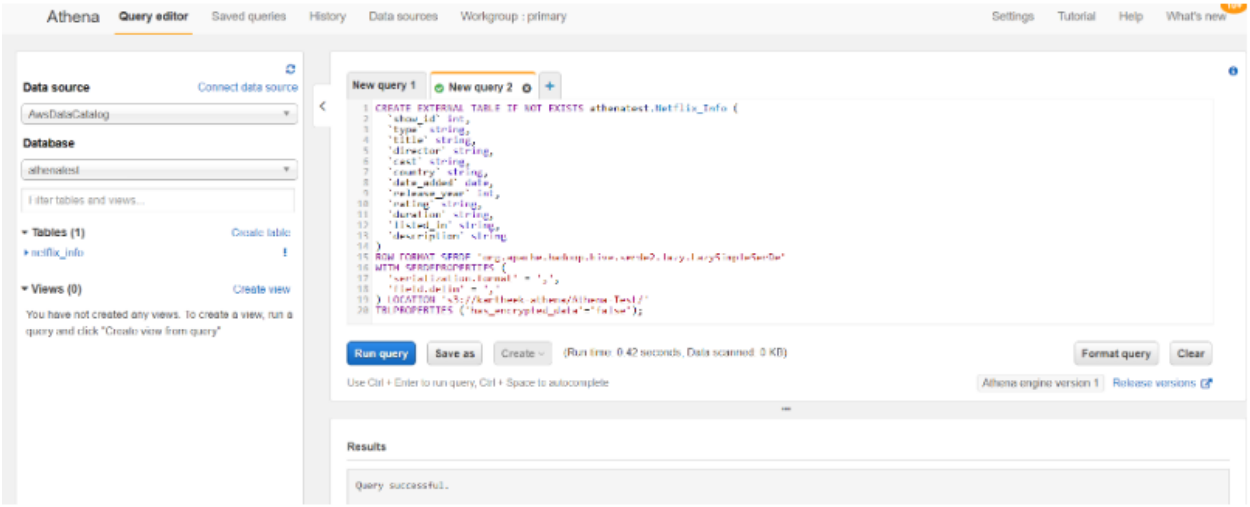


Add columns for the table, in order to add multiple columns at a time, we can click on the **bulk add columns** .

Define schema for the table and click on the next button.



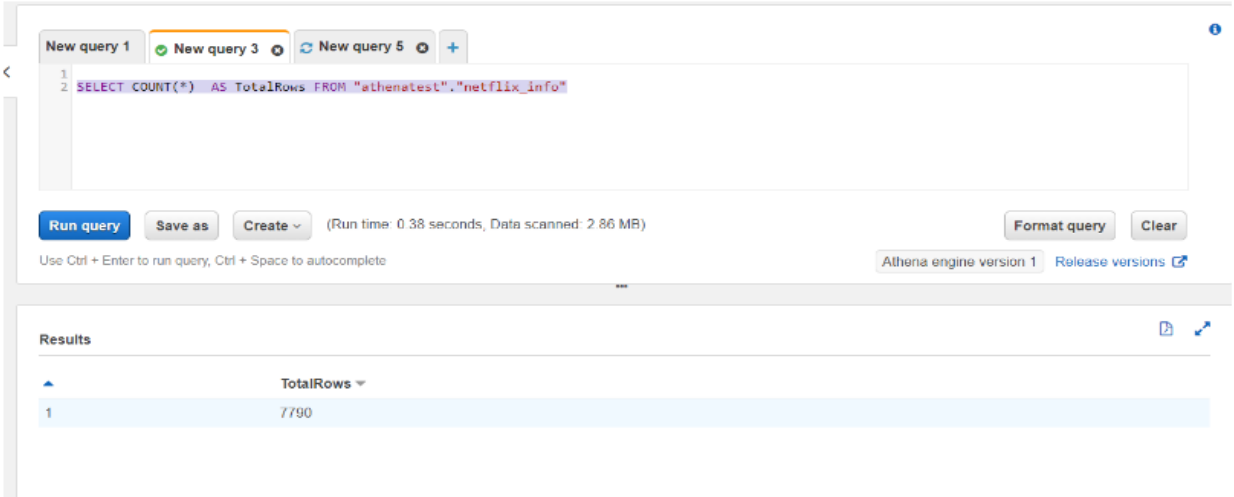
- Partitions are the way to group specific information and it is option and we can proceed further by clicking on the **Create table** button.
- It will redirects us to the **Query Editor** screen



- Now we can start querying data by writing simple queries.

```
SELECT COUNT(*)  AS TotalRows FROM "athenatest"."netflix_info"
```

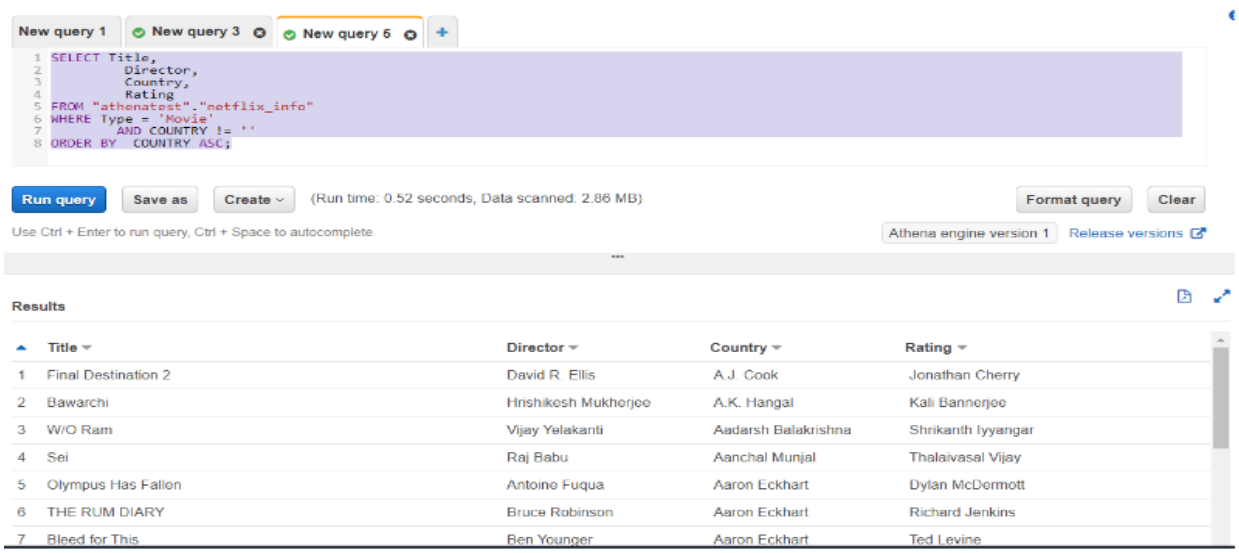
Result



- **Query2**

```
SELECT Title,
       Director,
       Country,
       Rating
FROM "athenatest"."netflix_info"
WHERE Type = 'Movie'
      AND COUNTRY != ''
ORDER BY  COUNTRY ASC;
```

Result



By this way we can make use of SQL to Query S3 files with **AWS Athena**

Week-13	Create and Configure Virtual Machines using Azure

Pre-Lab Questions	1. What is a Virtual Machine? When should you use Virtual Machine in Azure? 2. What are the various components of Virtual Machine? 3. What is a disk? 4. Is it possible to expand the size of OS Disk?

Virtual Machine

- It is a web service that provides secure, scalable compute capacity in the cloud. It is a virtual server in Azure virtual machine for running applications on the Azure infrastructure.
- Azure currently supports a variety of operating systems including: Ubuntu, Windows Server, Red Hat Enterprise Linux 7.1, CentOS, Debian and Oracle Linux

Steps to Create Windows Virtual Machine and Connect using Remote Desktop Connection:

- Login with Azure Account
- Go to Virtual Machine
- Click on Create
- Click on virtual machine
- Create Resource group
- A resource group is a container that holds related resources for an Azure solution. The resource group can include all the resources for the solution, or only those resources that you want to manage as a group.
- Resources can only exist in our resource group
- Resource group cannot be renamed
- Groups can have resources of many different types (Services)
- Groups can have resources from many different regions
- Click on create new
- Enter the name of resource group
- Click on ok
- Enter the name of the virtual machine
- Select the region

- Click on All images for selecting operating system
- Click on See all sizes
- Select B1s (Free Service eligible)
- Click on Select.
- Enter the username & password.
- Rules for Password:
- Password must have 3 of the following:

  1 lower case character,

  1 upper case character,

  1 number &

  1 special character

  The value must be between 12 and 123 character long.
- Enable RDP Port
- Note: For Windows Instance RDP Port needs to be open & for Linux Instance SSH Port
- Click on Next: Disks
- Click on Next: Networking
- Click on Next: management
- Click on Next: Advanced
- Click on Next: Tags
- Click on Next: Review + Create
- Click on Create
- Click on Name of the VM
- Copy the public IP Address
- Open Remote desktop Connection
- Enter the Public IP & Click on Connect
- Enter the username & password

Stop / Terminate the Instance:

- Select the instance
- Click on Delete / Stop

- Types Yes
- Click on Delete

Steps to Create Linux Virtual Machine and Connect using Putty:

- Go to Virtual Machine
- Click on Create
- Click on Virtual machine
- Select the resource group
- Enter the virtual machine name
- Select region
- Select the image (Ubuntu Server)
- Select the Size (B1s)
- Select Authentication type as SSH Public Key
- Enter Username
- Enter the key pair name
- Enable SSH Port
- Note: For Windows Instance RDP Port needs to be open & for Linux Instance SSH Port
- Click on Next: Disks
- Click on Next: Networking
- Click on Next: Management
- Click on Next: Advanced
- Click on Next: Tags
- Click on Next: Review + create
- Click on Create
- Click on Download Private Key and create resource
- Download Putty & Puttygen
- Open Website:
  https://www.chiark.greenend.org.uk/~sgtatham/putt y/latest.html
- Generate PPK File Using Puttygen
- Note down the public IP of our Linux Virtual Machine

- Open Putty
- Enter the public IP
- Attach PPK file with putty
- Click on open
- Click on accept
- Enter the username
- Press Enter

Stop / Terminate the Virtual Machine:

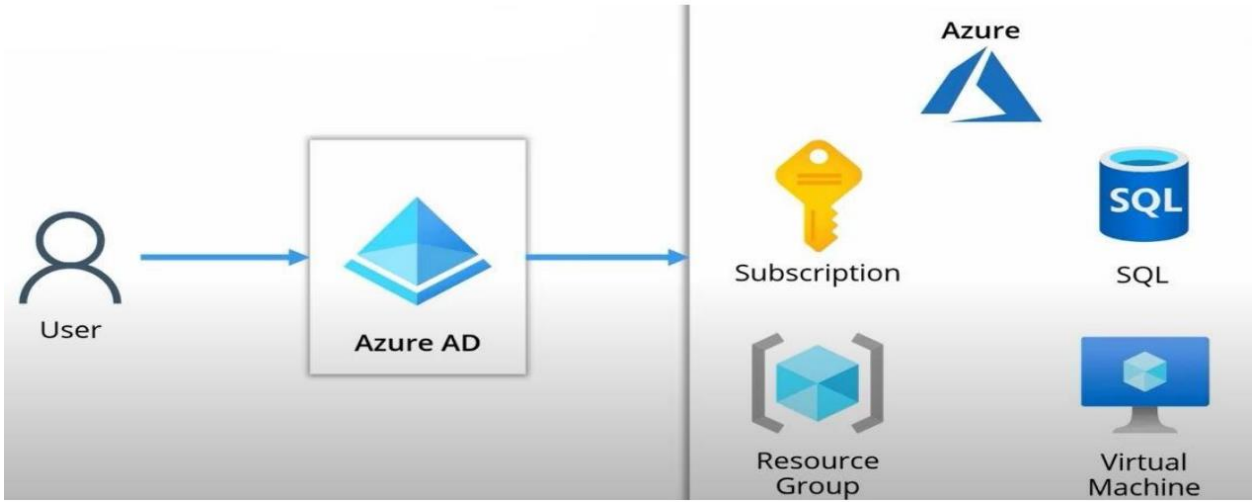
- Select the virtual machine
- Click on Stop / Delete
- Type as Yes
- Click on Delete

Week-14	**Create and Configure Virtual Machines using Azure** Create and manage users and groups in Azure AD, including user attributes, group memberships, and roles.

Pre-Lab Questions	1. What is a Virtual Machine? When should you use Virtual Machine in Azure? 2. What are the various components of Virtual Machine? 3. What is a disk? 4. Is it possible to expand the size of OS Disk?

Azure Active Directory:

Azure Active Directory is Microsoft's multi-tenant, cloud-based directory and identity management service. For an organization, Azure AD helps employees sign up to multiple services and access them anywhere over the cloud with a single set of login credentials.



Who uses Azure AD?

IT admins: As an IT admin, use Azure AD to control access to your apps and your app resources, based on your business requirements. For example, you can use Azure AD to require multifactor authentication when accessing important organizational resources.

App developers: As an app developer, you can use Azure AD as a standards-based approach for adding single sign-on (SSO) to your app, allowing it to work with a user's pre-existing credentials.

Microsoft 365, Office 365, Azure, or Dynamics CRM Online subscribers: As a subscriber, you're already using Azure AD. Each Microsoft 365, Office 365, Azure, and Dynamics CRM Online

tenant is automatically an Azure AD tenant. You can immediately start to manage access to your integrated cloud apps.

Azure Active Directory – Benefits

- One place for identity and access management
- One identity for all applications
- Security
- Ease of use
- Collaboration

Azure Active Directory – Features

- Application management
- Authentication
- Business-to-business (B2B)
- Business-to-customer (B2C)
- Device management
- Hybrid identity
- Identity governance
- Identity protection
- Reports and monitoring

Azure Active Directory – Password

- Password must contain three out of four of the following:
- Lowercase characters
- Uppercase characters
- Numbers (0-9)
- Symbols

Azure Active Directory - Create User

- Search Azure active directory
- Click on users
- Click on add new user
- Select create user

- Enter the username
- Select let me create the password
- Enter the password
- Click on create
- Click on the name of the new user
- Note down the id
- Now go to login page of Azure
  https://portal.azure.com/ • Enter the ID & password

AD Group

You can create a basic group using the Azure Active Directory (Azure AD) portal. For the purposes of this article, a basic group is added to a single resource by the resource owner (administrator) and includes specific members (employees) that need to access that resource.

Group Types

Security: Used to manage member and computer access to shared resources for a group of users. For example, you can create a security group for a specific security policy.

Microsoft 365: Provides collaboration opportunities by giving members access to a shared mailbox, calendar, files, SharePoint site, and more. This option also lets you give people outside of your organization access to the group.

Membership Types

Assigned: Lets you add specific users to be members of this group and to have unique permissions

Dynamic user: Lets you use dynamic membership rules to automatically add and remove members. (Premium License Required)

Dynamic device: Lets you use dynamic group rules to automatically add and remove devices. (Premium License Required)

Steps to Create Azure Active Directory Groups:

- Go to Azure active directory
- Click on groups
- Click on new group

- Enter the group name
- Enter the group description
- Click on create
- Click on name of the group
- Click on members
- Click on add members
- Search your user
- Select the member
- Click on select

Week-15	Create, Configure and manage Azure SQL Database

Pre-Lab Questions	1. What is Azure SQL database? 2. How does Azure SQL differ from SQL Server? 3. What is an SQL pool in Azure? 4. What is Azure SQL data sync? 5. What is an elastic pool? 6. How does the Azure SQL database scale?

Azure Database for MySQL is a relational database service in the Microsoft cloud based on the MySQL Community Edition.

Azure Database for MySQL powered by the MySQL community edition is available in two deployment modes:

1. Flexible Server
2. Single Server

Steps to Connect MySQL Database using Azure:

- Search Azure Database for MySQL
- Click on create
- Select resource type as Flexible server
- Click on create
- Select the resource group
- Enter server name (Unique Name)
- Select the region (East US)
- Select the Version (5.7)
- Select workload type as For Development or hobby projects
- Select has automatically select the compute & storage or if we want to change click on configure server
- Change compute size to 400 Max iops
- If we want to set high availability then we need to check the box
- Backup option we can select as per our requirement
- Click on save

- Enter username & password
- Click on next: Networking
- Select the public access
- Click on Add 0.0.0.0 or we can add the specific IP from where we want to access the database.
- Click on continue
- Click on Review + create
- Click on create
- Click on the name of the database
- Click on Backup & restore
- Click on restore for restore the database.
- Enter the details as per the client requirement
- Replication
- Click on replication
- Make sure you change the Compute to general purpose or system will give you the error message.
- Click on Add replica
- Enter the name of the replica
- Select the location
- Click on ok

Note: Read replicas are created with the same server configuration as the master. The replica server configuration can be changed after it has been created. The replica server is always created in the same resource group and same subscription as the source server. If you want to create a replica server to a different resource group or different subscription, you can move the replica server after creation. It is recommended that the replica server's configuration should be kept at equal or greater values than the source to ensure the replica is able to keep up with the master

- Download MYSQL Workbench: Click Here
  MySQL :: Download MySQL Workbench

- Note: To Download Workbench for windows Microsoft Visual C++ 2019 is mandatory in our system.

  Latest supported Visual C++ Redistributable downloads | Microsoft Learn
- Click on the overview
- Note down the server name & server admin login name
- Open MySQL Workbench
- Click on Database
- Click on Connect to database
- Enter the host name (Server Name)
- Enter the username (server admin login name)
- Click on store in Vault
- Enter the password
- Click on ok
- Click on ok
- Now we are able to connect to the database.
- SQL Query:

```
-- Create a database
-- DROP DATABASE IF EXISTS quickstartdb;
CREATE DATABASE quickstartdb;
USE quickstartdb;
-- Create a table and insert rows
DROP TABLE IF EXISTS inventory;
CREATE TABLE inventory (id serial PRIMARY KEY, name VARCHAR(50), quantity INTEGER);
INSERT INTO inventory (name, quantity) VALUES ('banana', 150);
INSERT INTO inventory (name, quantity) VALUES ('orange', 154);
INSERT INTO inventory (name, quantity) VALUES ('apple', 100);
-- Read
```

```
SELECT * FROM inventory;
-- Update
UPDATE inventory SET quantity = 200 WHERE id = 1;
SELECT * FROM inventory;
-- Delete
DELETE FROM inventory WHERE id = 2;
SELECT * FROM inventory;
```